

Christopher Liu *Senior Data Engineer*

✉ liuc.dev@gmail.com ☎ (669) 600-0330 📍 Campbell, CA 🔗 [linkedin.com/in/chrissjliu](https://www.linkedin.com/in/chrissjliu)

PROFILE

Senior Data Engineer with 9+ years of experience building scalable, reliable, and high-performance data platforms across social media, creator economy, and e-commerce domains. Skilled in developing real-time and batch data pipelines using Apache Spark, Databricks, Snowflake, Airflow, Kafka, and modern cloud ecosystems including AWS, GCP, and Azure. Proven track record optimizing data infrastructure to reduce latency, improve cost efficiency, and strengthen data quality and observability. Passionate about enabling experimentation, analytics, and ML workflows that drive impactful, data-informed product decisions.

SKILLS

Language — Python, SQL, Java, PySpark, Scala, Bash

Pipeline & Orchestration — Apache Airflow, dbt, Apache Spark, Delta Lake, Kafka

Warehousing & Lakehouse — Snowflake, Databricks, Redshift, BigQuery

Cloud Ecosystems — AWS, Azure, GCP

BI & Visualization — Looker, Tableau, Power BI, Metabase

AI & ML Pipelines — MLflow, HuggingFace Transformers, Databricks ML

Graph & NoSQL Technologies — Neo4j, AWS Neptune, DynamoDB, ClickHouse

Versioning & Testing — GitHub, dbt Cloud, CI/CD, dbt tests, observability tools

PROFESSIONAL EXPERIENCE

Discord

01/2023 — Present

Senior Data Engineer

- Built and maintained monetization data pipelines using Airflow, Kafka, and Python, ingesting subscription lifecycle and billing data into Snowflake via Snowpipe for use in LTV, churn, and revenue reporting.
- Tuned Snowflake performance through optimized clustering, partitioning, and compute resource allocation, reducing dashboard latency by 45% and improving warehouse cost efficiency.
- Supported a cloud-native architecture across AWS and GCP, collaborating with platform engineers and DevOps to integrate CI/CD workflows, observability tooling, and cross-cloud data access policies.
- Built a scalable experimentation insights platform using Java, Databricks Workflows, PySpark, and Delta Lake to aggregate A/B test metrics across platform, geography, and user segments.
- Modeled bronze-silver-gold Delta tables with dbt and implemented statistical significance logic in Python, partnering with data scientists to align on methodology and ensure statistical rigor.
- Delivered experiment dashboards in Looker and Tableau, working with product managers and UX researchers to translate test results into actionable insights for rapid iteration and feature rollouts.
- Developed modular data models in dbt to transform subscription events into gold-layer tables for Finance, and Product teams, collaborating with analysts and PMs to define business logic and KPIs.
- Designed and implemented data quality monitoring using dbt tests, Great Expectations, and Airflow, helping analytics and BI teams proactively catch schema drift, null spikes, and broken reports.

Meta

04/2021 – 12/2022

Data Engineer

- Engineered large-scale ETL pipelines using Hive, Presto, Snowflake, Airflow/Dataswarm, Python, SQL to process billions of daily events and deliver timely, reliable engagement metrics.
- Designed and deployed a unified cross-platform data model using Hive Metastore, Presto SQL, Spark SQL, iData to reduce data redundancy by 20+ TB and improve metric accuracy across analytics teams.
- Implemented automated data-quality validation and anomaly-detection workflows using Dataswarm operators, Python, SQL validators, CDM monitoring to reduce data incidents by 30 percent and increase dataset reliability.
- Built real-time analytics dashboards using Scuba, streaming log pipelines, Scribe or Kafka-like internal systems, Kibana-style visualizations to surface feature performance within minutes.
- Collaborated with data scientists, product managers, and software engineers to define logging requirements and deliver clean datasets using Python, Hive, Spark, Thrift or Protobuf logging schemas to support A/B testing for 100M+ users.
- Optimized multi-terabyte Spark pipelines by tuning Java-based Spark operators, pushing predicate filters down, rewriting skewed joins, and adjusting JVM memory parameters to cut compute cost 25% and improve runtimes 40%.

Minted

05/2017 – 04/2021

Data Engineer

- Developed and maintained ETL pipelines using Airflow, Python, and AWS Redshift to automate data ingestion from multiple sources, improving data freshness and reliability for analytics teams.
- Contributed to the design and implementation of a centralized data warehouse, optimizing data models for reporting and enabling faster, more consistent insights across marketing, finance, and product teams.
- Collaborated with analysts to deploy Looker dashboards and models (LookML + dbt), improving accessibility to metrics and reducing dependency on manual SQL queries by business users.
- Supported development of real-time data processing pipelines using Kafka and Spark Streaming to capture key site events and provide near-real-time operational metrics.
- Partnered with data scientists to prepare and process training data for recommendation models, helping launch personalized product recommendations that improved engagement and conversion rates.
- Improved data quality and monitoring by implementing validation checks, logging, and alerting in Airflow, reducing data errors and improving trust in analytics deliverables.

Minted

08/2016 – 03/2017

*Data Engineer Intern***EDUCATION**

University of Southern California

08/2011 – 05/2015

Bachelor's Degree - Computer Science